# Predict a position and a range

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Predict a future position and explain the limits of the prediction.

**SUCCESS CRITERIA - I can...**

1. Use matching units and include the starting position.
2. Report a central prediction and its bounds.
3. Explain how an uncertain measurement differs from a limit of the calculation.

**Core:** teacher selects from 1-9. **Stretch:** 10 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

Start at **5 m**, move to **14 m**, then finish at **7 m**.
1. **Draw the route**.
2. **Find distance**, displacement with direction and final position.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** $x_f = x_i + v\,\Delta t$.
Constant velocity; i = initial, f = final. Use the same units for all positions.

**Worked reasoning:** A cart starts at **1 m** and moves right at **2 m/s** for **3 s**. Central prediction: **1**+**2**×**3**=**7 m**. If its speed could be **1** to **3 m/s**, the endpoint range is **4** to **10 m**.

**Your turn:** A cart moves along a marked track.
1. **Show its start**, central predicted endpoint, and full possible endpoint range on the number line.
2. What must remain constant throughout the **3 s**?

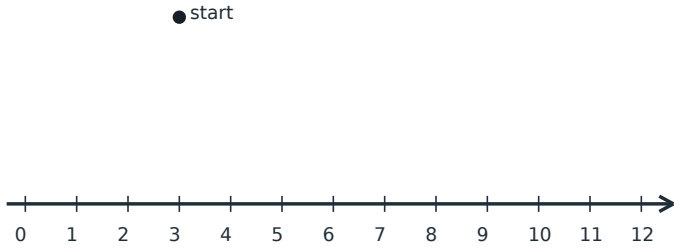



Position (m); right is positive

**Marzano check - circle:** ① ② ③

## 03 Start plus change

A cart starts at **3 m** and moves right at **2 m/s** for **4 s**. **Find its displacement** and final position. **Mark the endpoint**.




Position (m); right is positive

**Marzano check - circle:** ① ② ③

## 04 Predict leftward motion

A robot starts at the **12 m** mark and drives left at a steady **2 m/s** for **3 s**.
1. **Draw the move** on the number line.
2. **Find the displacement** and final position.




Position (m); right is positive

**Marzano check - circle:** ① ② ③

# Predict a position and a range

**antocciphysics**

Continue down each column.

### 05 Match the time units

A rover moves at **2 m/s** for **1 minute**.
1. Use **1 minute** = **60 seconds**.
2. **Find its distance** at this steady speed.

**Marzano check - circle:** ① ② ③

### 06 Read the range

A prediction is **10 m** with a range from **8 m** to **12 m**.
1. **Label the middle**, lower end, and upper end.
2. **Find how far each end** is from the middle.

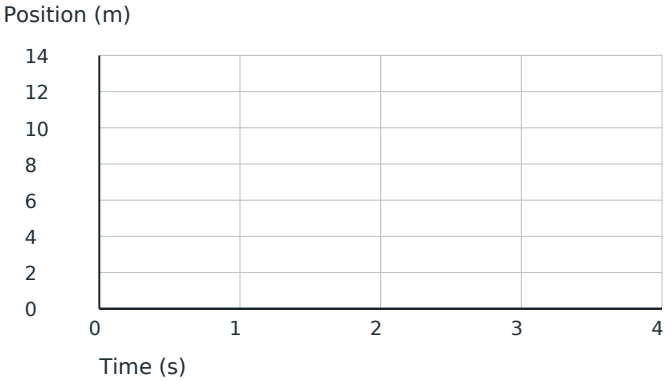



**Marzano check - circle:** ① ② ③

### 07 A slow and a fast case

A rover starts at **0 m** and travels right at a constant speed somewhere between **1** and **3 m/s**.
1. **Draw the slowest** and fastest possible position-time lines.
2. **Find the possible position range** after **4 s**.

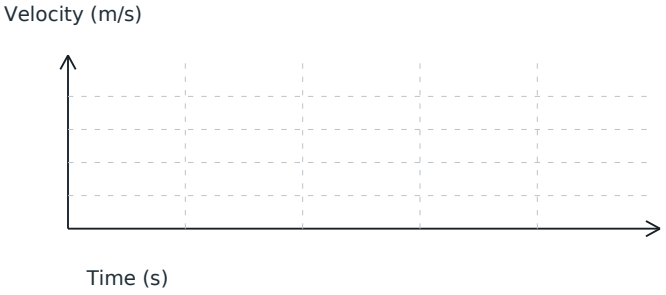
**Support:** Slow case: ____. Fast case: ____. My range is ____.




**Marzano check - circle:** ① ② ③

### 08 Review: connect an earlier idea

A velocity graph is constant at **-3 m/s** for **5 s**. **Find displacement**. A second object has components **7 m/s** *east* and **2 m/s** *south*; **sketch its direction**. **Sketch the velocity**–time graph; **label the given values**.




**Marzano check - circle:** ① ② ③

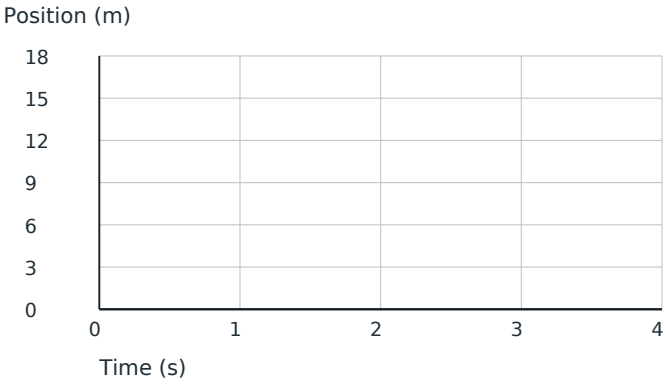
### 09 Exit check

A cart starts at **2 m** and moves right at **3 m/s** for **2 s**. **Find the central final position**. If the possible final positions run from **7** to **9 m**, state the range and one thing you are treating as true.

**Marzano check - circle:** ① ② ③

### 10 STRETCH: test the prediction

1. Design two different starting-position and speed pairs that both give an endpoint of **10 m** after **2 s**.
2. **Draw the routes**.




**Marzano check - circle:** ① ② ③